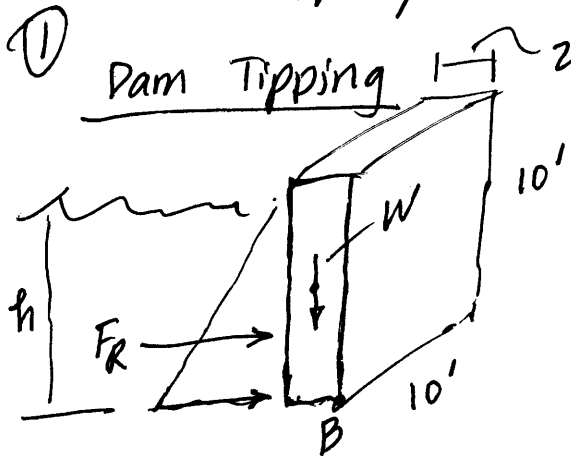


Fluid Pressure Overturning, Gate Opening Probs

Some fluid pressure problems involve a dam tipping or a gate opening.



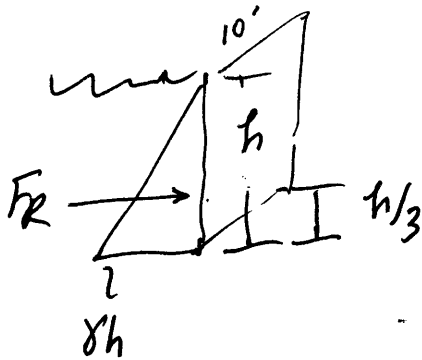
concrete dam.

$$\text{Weight } W = (10)(10)(2)(150 \text{ lb/ft}^3) \\ = 30,000 \text{ lb}$$

at what depth of water
will this dam or wall tip?

(assume no ship)

$$\gamma = 62.4 \text{ lb/ft}^3$$



$$F_R = \frac{1}{2} b h w = \frac{1}{2} (8h) (h) (10)$$

acts at $h/3$

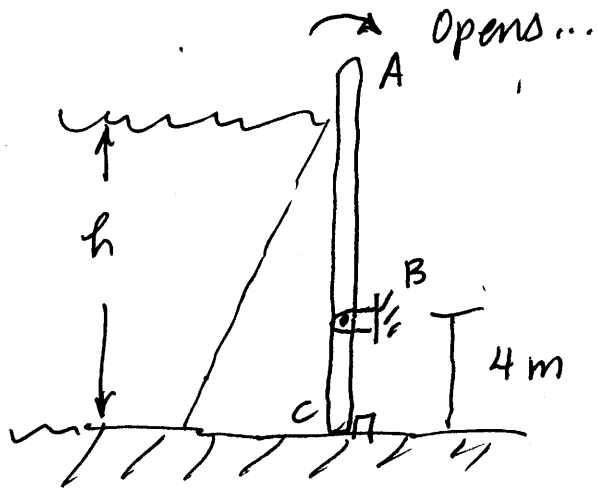
$$+ \sum M_B = 0; \quad F_R \left(\frac{h}{3} \right) - 30,000 (1 \text{ ft}) = 0$$

$$(5)(62.4) h^2 \left(\frac{h}{3} \right) = 30,000$$

$$h^3 = 288.5$$

$$\boxed{h = 6.607 \text{ ft}}$$

② Gate Opening



at what height, h ,
of water will
gate ABC open?
(Has a pivot or hinge
at B, and a
stop at C.)